



Exploring & Describing Data

Topic Test 1 Solution

Total Marks: 30

Time: 50 minutes

- 1 The test scores of a Mathematics class and their frequencies are shown in the table. Which of the following measures will change if another score of 60 is added to the dataset? 1

A. Mean

B. Range

C. Median

D. Mode

Score	Frequency
52	4
54	8
56	12
58	2
60	2

- 2 The stem-and-leaf plot shows the number of goals scored by a team in each of 10 netball games. Find the standard deviation for this data set, correct to the nearest whole number? 9

0	6	8		
1	2	4	5	
2	1	5	5	9
3	5			

- 3 The number of peas in each pod of a large sample of a particular species of pea are normally distributed and shown in the table below.

Number of peas/pod	Number of pods
2	10
3	16
4	38
5	71
6	80
7	73
8	35
9	19
10	8
11	0
12	1

- a. Determine the mean of the distribution. Give your answer correct to the nearest number of peas/pod. 6

- b. Find the standard deviation. Give your answer correct to three significant figures. 1.75

- c. Determine by calculation if 12 peas/pod is an outlier. 3

$$\begin{aligned}
 Q_1 &= 5 & Q_3 &= 7 \\
 \therefore IQR &= 2 \\
 Q_3 + 1.5 \times IQR \\
 &= 7 + 1.5 \times 2 = 10 \\
 \therefore 12 \text{ is an outlier}
 \end{aligned}$$

- 4 A golfer's scores for the first nine holes of a golf course are: 4 3 5 6 4 3 8 6 6

3 3 4 4 5 6 6 6 8

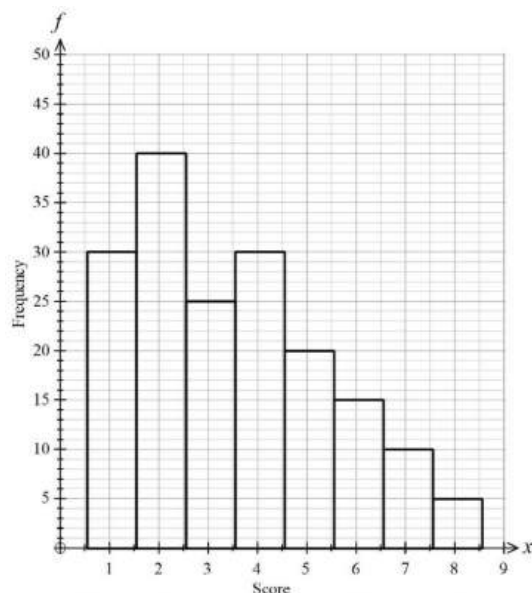
- a. Calculate the range of the data set. 8 - 3 = 5

- b. Find the median of the data set. 5

- 5 The frequency histogram below displays the scores on a test. What was the mean score on the test?

1

$$\begin{aligned} \text{mean} &= \frac{30 \times 1 + 40 \times 2 + 25 \times 3 + 30 \times 4 + 20 \times 5 + 15 \times 6 + 10 \times 7 + 5 \times 8}{175} \\ &= 3.46 \end{aligned}$$



- 6 Data was collected from 30 students on the number of text messages they had sent in the previous 24 hours. The set of data collected is displayed below. Determine whether the data has an outlier. Justify your answer with calculations.

2

Male		Female
9 9 8 7 6 5 5 4 2 1	0	8 9
1 (1) 0 0	1	(1) 1 2 5 6 8 8 8
15 th 0	2	16 th 0 1 7
	3	4
	4	
	5	
	6	
1	7	

$$\begin{aligned} \text{Median} &= \frac{15^{\text{th}} + 16^{\text{th}} \text{ score}}{2} = 11 \\ Q_1 &= 8^{\text{th}} \text{ score} = 8 \\ Q_3 &= 22^{\text{nd}} \text{ score} = 18 \\ IQR &= 18 - 8 = 10 \\ \text{Outlier} &\rightarrow Q_3 + 1.5 IQR = 18 + 1.5 \times 10 = 33 \\ \therefore 71 &\text{ is an outlier} \end{aligned}$$

- 7 The heights, in metres, of 9 buildings in a small tourist area are as shown.

3

50, 54, 56, 58, 58, 68, 70, 74, 95

Is the height of the tallest building in this area considered an outlier? Justify your answer with calculations.

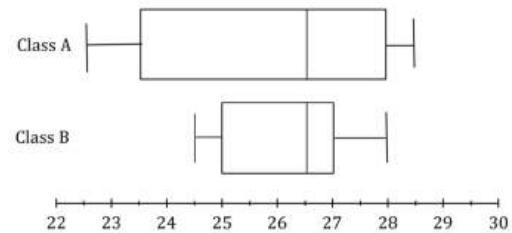
$$\begin{aligned} Q_1 &= 55, \quad Q_3 = 72 \\ IQR &= 72 - 55 \\ &= 17 \\ \text{outlier} &> 1.5 \times 17 + Q_3 \\ \text{outlier} &> 1.5 \times 17 + 72 \\ &> 97.5 \\ \therefore 95 &\text{ is not an outlier.} \end{aligned}$$

- 8 Two school classes performed an experiment to determine the correct volume of a chemical solution. Their results, in millilitres, were presented in the parallel box-plot below.

a. What percentage of the results found by

Class A were below 28 mL?

Below Q3 therefore 75%



b. Find the Range and Interquartile range for the results found by each class.

Class A

$$\begin{aligned} IQR &= Q_3 - Q_1 \\ &= 28.0 - 23.5 = 4.5 \\ \text{Range} &= 28.5 - 22.5 = 6 \end{aligned}$$

Class B

$$\begin{aligned} IQR &= Q_3 - Q_1 \\ &= 27.0 - 25 = 2 \\ \text{Range} &= 28.0 - 24.5 = 3.5 \end{aligned}$$

c. One student claimed that there was no difference between the results of the classes. Is

the student's comment reasonable if based on the data? Using the box plots, compare and contrast the data to justify your response.

The medians are equal and therefore it is reasonable for the student to assert that there appears to be no difference in the volume results obtained from both classes. (and/or)

However, the results of Class A indicate a larger interquartile range and range. This indicates the results of class A are less consistent than class B.

- 9 On a rail station, a train is scheduled to arrive at 9:00am each morning. The following data shows the times the train arrived on seven consecutive days.

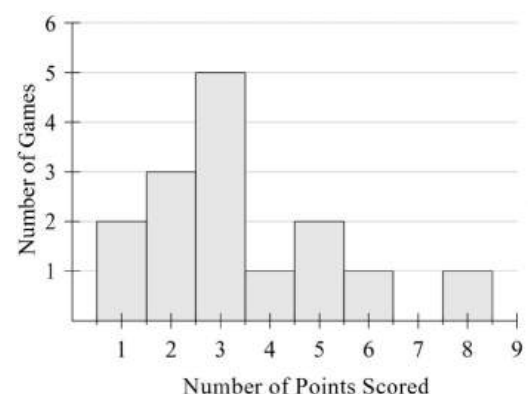
9:03am, 9:04am, 9:02am, 9:11am, 9:02am, 9:03am, 9:01am

With the aid of calculation, determine whether the arrival time of 9:11am could be considered an outlier for the presented data.

9:01 (9:02) 9:02 (9:03) 9:03 (9:04) 9:11
 $IQR = 2 \text{ mins.}$
 $9:04 + 1.5 \times 2 = 9:07$
 ∴ yes, 9:11 is an outlier as it is $> 9:07$

- 10 A sports team recorded the number of points scored each game. The frequency histogram shows the number of points scored each game across 15 games. What is the mean number of points scored per game for this team?

3.4



- 11 The stem and leaf plot shows the number of shots needed to complete a game of pool on two tables.

Twenty games were played on each table. Is the result of 69 shots an outlier for either of the tables?

Table A	Table B
7 6 4	6 8
8 4 4 1	2 6 7 7
8 8 7 7 5 3	0 2 3 4 4
6 5 5 4	1 3 5
2 1	5 5 6 8 9
	5
	6 9
Q ₁ =14 Q ₂ =27 Q ₃ =35	Q ₁ =17 Q ₂ =24 Q ₃ =45

$$IQR = 35 - 14 = 21$$

$$Q_3 + 1.5 \times IQR = 35 + 1.5 \times 21 = 66.5$$

69 is greater than 66.5 so outlier

$$IQR = 45 - 17 = 28$$

$$Q_3 + 1.5 \times IQR = 45 + 1.5 \times 28 = 87$$

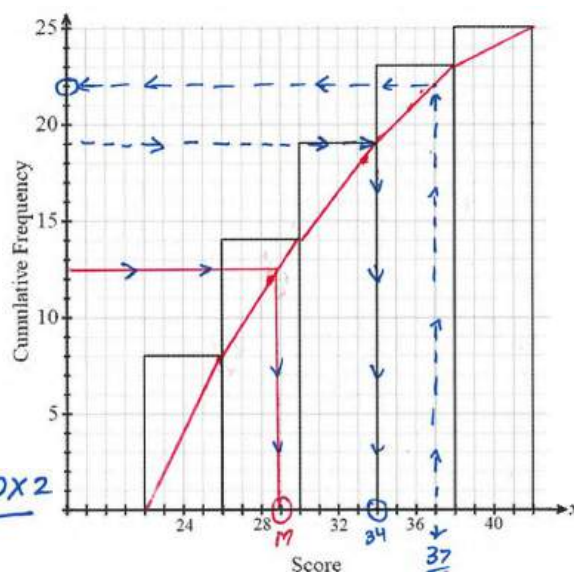
69 is less than 87 so not an outlier

Yes, an outlier for Table A only

- 12 Sam recorded the scores of 25 footballers who each took 50 shots at goal. The cumulative frequency graph displays the results.

- Use the graph to estimate the median number of goals scored.
- Calculate the mean number of goals scored. (Answer to 1 decimal place.)

$$\bar{x} = \frac{24 \times 8 + 28 \times 6 + 32 \times 5 + 36 \times 4 + 40 \times 2}{25} = 29.8$$



- What percentage of players scored 37 goals or more?

3 students scored 37 or more.
 $\frac{3}{25} = 12\%$

- The players with the top 76% of scores go through to the next round of shots. What score was needed to go through to the next round?

$$76\% = \frac{19}{25}$$

all players with 34 or more goals move to the next round of shots.
 $\therefore$ 34 or more goals.



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Topic Test 2 Solution

Total Marks: 30

Time: 50 minutes

- 1 Mr Miller gave his Year 7 Mathematics class a topic test. On the first attempt, the class had a mean mark of 65 and a standard deviation of 5. The class was given a chance to sit the test again and every student improved by exactly 20 marks. What would happen to the class mean and standard deviation? The mean increased and the standard deviation remained the same. 1

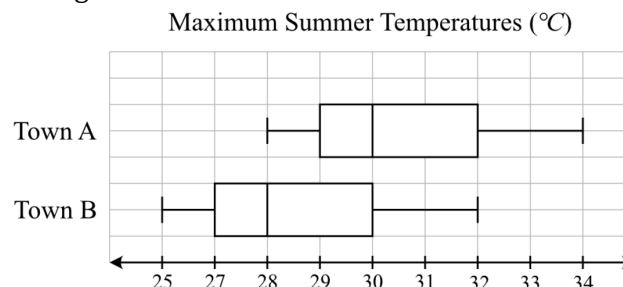
- 2 Mr Johnson gave a maths test. The results are listed. What is the difference in the median for the females and males is? 2

Females			Males
9 7	0	9	
8 8 6 4 3	1	2 3 5 7 8	
8 7 4 2	2	1 1 6 6 7 9	
8 7 4 3 1	3	1 2 3	

- 3 A set of data has a lower quartile of 36 and an upper quartile of 48. What is the maximum possible range if there are no outliers in the data set? 3

$$\begin{aligned}
 Q_1 &= 36 & Q_3 &= 48 \\
 IQR &= Q_3 - Q_1 = 48 - 36 = 12 \\
 \text{Outlier lower bound} &= Q_1 - 1.5 \times IQR \\
 &= 36 - 1.5 \times 12 = 18 \\
 \text{Outlier upper bound} &= Q_3 + 1.5 \times IQR \\
 &= 48 + 1.5 \times 12 = 66 \\
 \text{max. possible range} &= 66 - 18 = 48
 \end{aligned}$$

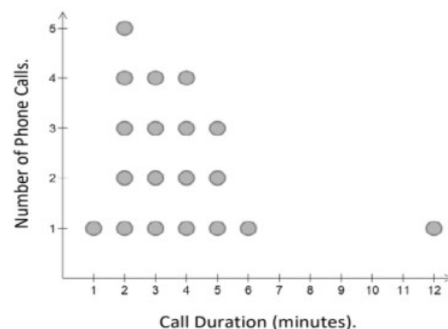
- 4 In one particular Australian summer, the summer months of December, January and February had a total of 90 days. The box plots below display data on the maximum daily temperatures in town A and town B during these summer months.



- a. What is the median maximum summer temperature of Town A? 30°C 1
- b. What percentage of summer days in Town A were warmer than Town B's warmest day? 1
- c. How many summer days in Town B were at least 28°C ? 25% 2

$$\begin{aligned}
 &28^{\circ}\text{C is the median for Town B} \\
 &50\% \text{ of temperatures are greater than } 28^{\circ}\text{C} \\
 &50\% \text{ of } 90 = 45 \text{ days}
 \end{aligned}$$

- 5 David recorded the duration of his phone calls (in minutes) over a day. The results are shown on the dot plot.



- a. What was the interquartile range of David's calls?

$$\begin{aligned} Q_2 &= 3 \\ Q_1 &= 2 \\ Q_3 &= 5 \end{aligned} \quad \checkmark \quad IQR = 5 - 2 = 3$$

- b. Justify, using calculations, that David's twelve-minute call was an outlier.

$$Q_3 + 1.5 \times IQR = 5 + 1.5 \times 3 = 9.5$$

12 is an outlier

- 6 John and Harry record the number of sales they made over the past two weeks. The results are recorded in two separate stem and leaf plots. Find the difference in the interquartile ranges between John and Harry.

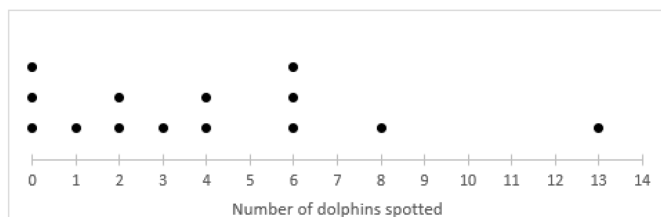
John		Harry	
Stem	Leaf	Stem	Leaf
4	2 3	3	2 3 3 8
5	1 1 4 6 6	4	4 5 5 7
6	2 3 3 7 7 8	5	2 4 6 6
7	3 4 4	6	4 7 7
		7	2 3
		8	4 4

$$IQR \text{ for John} = 67.5 - 52.5 = 15$$

$$IQR \text{ for Harry} = 67 - 44 = 23$$

$$\text{Difference} = 8$$

- 7 The dot plot below shows the number of dolphins spotted by passengers on a cruise ship each day over a two-week period. Determine if there are any outliers in the data. Justify your answer with calculations.



$$\text{Number of scores} = 14$$

$$\text{Median} = 4 \text{ (Point between 7th and 8th score)}$$

$$Q_1 = 1 \text{ (4th score)}$$

$$Q_3 = 6 \text{ (11th score)}$$

$$IQR = Q_3 - Q_1$$

$$IQR = 5$$

$$\text{Lower outlier limit} = Q_1 - 1.5 \times IQR$$

$$\text{Lower outlier limit} = 1 - 1.5 \times 5$$

$$\text{Lower outlier limit} = -6.5$$

No lower outlier

$$\text{Upper outlier limit} = Q_3 + 1.5 \times IQR$$

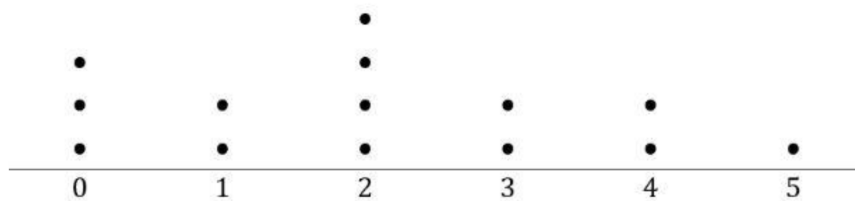
$$\text{Upper outlier limit} = 6 + 1.5 \times 5$$

$$\text{Upper outlier limit} = 13.5$$

No upper outlier

There are no outliers in this data as all scores are between -6.5 and 13.5

- 8 14 people were asked to indicate the time (in hours) they had spent watching television on the previous night. The results are displayed in the dot plot below. Find the mean and population standard deviation of these times. Correct your answers to one decimal place. 2



Using a calculator,

$$\bar{x} = \frac{0 + 0 + 0 + 1 + 1 + 2 + 2 + 2 + 2 + 3 + \dots}{14} = 2.1$$

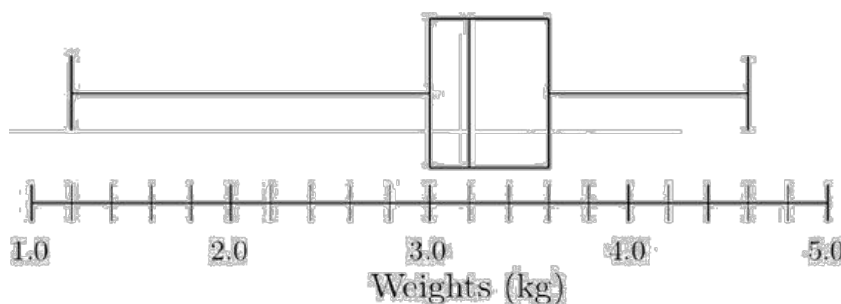
$$s = 1.5$$

- 9 19 people had their heart rate measured (in beats per minute) both before and after an exercise class.

Heart Rate	
Before Exercise	After Exercise
9 8 8 8	6
8 6 6 4 1 1 0	7
8 8 6 2	8 6 7 8 8
6 0	9 0 2 2 4 5 8 9 9
4	10 0 4 4
0	11 8
	12 4 4
	13
	14 6

- a. By how many beats per minute did the median heart rate increase after exercise? $98 - 76 = 22$ 1
- b. Calculate the interquartile range after exercise. 14 1
- c. Is 146 an outlier? Show by mathematical calculation. $104 + 1.5 \times 14 = 125$
 $\therefore 146$ is an outlier 2
- d. Describe the skewness of the data after exercise (positive, negative or symmetrical). positive skew 1

- 10 The birth weight of babies born in Australia during 2019 is presented in the following box-and-whisker diagram.



- a. What is the weight of the heaviest baby born in 2019? 4.6 kg 1
- b. Find the interquartile range. $3.6 - 3.0 = 0.6 \text{ kg}$ 1
- c. Calculate the upper limit for outliers in this data set. $Q_3 + 1.5(IQR) = 3.6 + 1.5(0.6) = 4.5 \text{ kg}$ 2
- d. Describe the shape of the distribution of weights. Negatively skewed 1